

Placebo effects: Rethinking the mechanisms of morphine analgesia

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A new study provides the most comprehensive, mechanistic account to date of how the brain learns to predict and mimic drug effects. After associating morphine injections with a place context, the context alone elicits endogenous opioid release and analgesia via a classic prefrontal-brainstem pathway.

Opioids are among the oldest and best-studied drugs in the pharmacopeia of modern medicine. Because they are so familiar, it is tempting to think that we understand how they work. At the molecular level, we do know a great deal. But the G protein-coupled receptors activated by opioids are embedded in complex networks of neurons throughout the brain. At the systems level, the mechanisms of even the old ‘pain-killing’ standby morphine are largely terra incognita. One of the most important clues that systems thinking is important for understanding how drugs work is the placebo effect: suggestions that one is receiving a pain-relieving treatment, coupled with supporting experiences of relief, can activate endogenous opioid systems and relieve pain^{1–3}. Thus, the effects of drugs and the context in which they are given are entangled. In a new study in *Neuron*, Livrizzi, Chang-Weinberg *et al.*⁴ dissect the circuits underlying placebo analgesia in mice, providing definitive evidence that they activate — and require — brainstem pathways crucial for many forms of pain control, including morphine.

Given that placebo effects typically involve verbal suggestions, can they be studied in rodents? They can: the solution involves classical conditioning, training the animals to associate drug delivery with other sensory cues — the experience of a drug injection itself, a unique taste, or place context cues⁵. This procedure, termed ‘pharmacological conditioning’, was pioneered in the early 20th century by Russian scientists — including Ivan Pavlov — studying brain control over autonomic, hormonal, and immune responses in the body⁶.

Livrizzi, Chang-Weinberg *et al.*⁴ used a pharmacological conditioning procedure

that paired drug delivery with a unique place context marked by visual and odor cues. Saline injections were repeatedly paired with one context, and morphine injections with another, over multiple days. Both contexts involved morphine or saline exposure followed by a hot-plate challenge, providing a measure of pain-related behavioral withdrawal. They found that after conditioning, exposure to the context alone resulted in robust analgesia — about 50% as large as the morphine effect — that was blocked by the opioid antagonist naloxone. Naloxone reversibility is a key feature of human placebo analgesia. It provided a first clue that contextual information could activate endogenous opioids, putting placebo effects ‘on the map’ as a neurobiological phenomenon⁷.

Livrizzi, Chang-Weinberg *et al.*’s study includes multiple built-in replications, including generalization across several pain assays: withdrawal from the hot plate, escape behavior, and mechanical sensitivity. They also showed transference across two types of conditioning, from pinprick to hot-plate and vice versa — another feature of human placebo analgesia⁸. Transfer across pain behaviors suggested that conditioning elicited analgesia rather than learned motor responses, an important distinction. Even more strikingly, morphine+hot-plate conditioning inoculated the animals against hypersensitivity produced by a common inflammatory pain model for multiple days — an effect also blocked by naloxone and therefore endogenous opioid-mediated.

These behavioral findings were accompanied by a dazzling array of techniques for dissecting the neural circuits that mediate them. The main focus was the periaqueductal gray (PAG),

a major center for opioid-mediated analgesia across species and a key node in descending modulatory control of spinal nociceptive afferents via the rostral ventral medulla (RVM) and other areas (Figure 1). Human studies have found placebo-induced opioid release in the PAG and associated medial (mPFC) and orbital prefrontal regions², and identified functional pathways connecting the mPFC, PAG, and spinal cord that mediate the effects of suggestions on pain⁹. However, only nonhuman studies can provide the precise control over this circuit required to both confirm its causal importance in pain control and establish systems-level and pharmacological mechanisms.

Livrizzi, Chang-Weinberg *et al.* used TRAP2 to label ensembles of neurons in the ventrolateral PAG (vIPAG) that were activated during conditioning, allowing them to record their activity and manipulate them chemogenetically. These ensembles included vIPAG→RVM projection neurons, and recording neural Ca²⁺ activity using fiber photometry revealed approximately equivalent responses in these descending projection neurons to both morphine and placebo. Chemogenetic re-activation of vIPAG→RVM ensembles produced thermal and mechanical analgesia, and silencing them completely abolished both morphine and conditioned placebo analgesia.

In addition to chemogenetic manipulation of the vIPAG→RVM pathway, new tools provided the first direct, temporally precise recording and manipulation of opioid activity during morphine and placebo analgesia. Second-by-second recording of endogenous opioid activity in μ -opioid receptor-expressing neurons using the genetically encoded opioid peptide

sensor δ Light (with Oprm1-Cre) shows placebo-induced opioid increases in the vIPAG within 10 seconds of mice being placed on the hot plate. Light-activated uncaging of naloxone in the vIPAG using photoactivatable naloxone (PhNX), an innovative recent technique, produces long-lasting reversal of morphine analgesia after uncaging, either before or after morphine administration. PhNX photoactivation in the vIPAG just prior to placing mice on the hot plate also completely reversed placebo analgesia. Together, these findings show that opioid signaling in the vIPAG $\rightarrow$ RVM circuit is both necessary and sufficient for both drug and placebo effects, providing evidence for a common pathway.

Pharmacological conditioning and other types of placebo effects rely on memory-driven conceptions of the context surrounding treatment. A crucial question is which regions store the context memories and translate them into activation of descending modulatory systems. Two key areas in humans are the lateral prefrontal cortex and mPFC — particularly anterior muncingulate (aMCC) and vmPFC — which show activation and increases in functional connectivity with PAG during placebo analgesia^{10,11}. Another is the insula, whose anterior portions in humans connect with mPFC to form a multi-modal system integrating affect, interoception, and cognitive control¹². This system is thought to support ‘cognitive maps’ that provide representations of the ‘self-in-context’, providing predictive control over both perception and action¹³.

A final key set of experiments in Livrizzi, Chang-Weinberg *et al.*'s study used rabies virus to identify monosynaptic inputs into vIPAG→RVM projection neurons. Labeled areas included ACC (Cg1), vmPFC (prelimbic and infralimbic) and anterior insula, along with somatomotor cortex and other areas. TRAP2 activation-based tagging with a retrograde AAV injected into the vIPAG showed that vIPAG-projecting neuronal ensembles in these areas were activated by the placebo context. Crucially, chemogenetic silencing of either mPFC→vIPAG or ACC→vIPAG projection neurons completely abolished placebo analgesia, but had no effect on morphine analgesia. Insula→vIPAG silencing had no effect on placebo or morphine analgesia. These results show

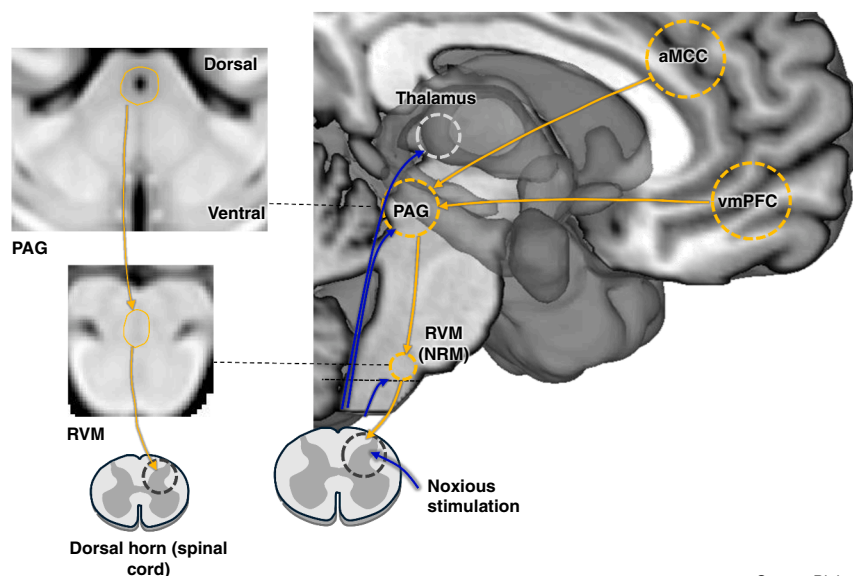


Figure 1. The descending prefrontal-PAG-RVM-spinal pathway.

Selected ascending (blue) and descending (gold) projections for transmission and regulation of nociceptive neurons in the spinal dorsal horn. The figure shows a rendition of the human brain, though Livrizzi, Chang-Weinberg *et al.* studied these regions in mice. They identified monosynaptic projections from the anterior cingulate cortex (ACC) and medial prefrontal cortex (infralimbic and prelimbic) to ventrolateral periaqueductal gray (vlPAG) that were activated on exposure to the placebo context (morphine-paired chamber) and necessary for placebo analgesia. The closest human homologues to these regions are likely the anterior midcingulate (aMCC) and ventromedial prefrontal cortex (vmPFC), respectively. The vmPFC is a broad zone covering multiple regions, including the pregenual and subgenual ACC. vlPAG projects to the rostral ventral medulla (RVM), which contains both 'on' (nociception-facilitatory) and 'off' (nociception-inhibitory) spinal projection neurons, providing bidirectional cerebral control over nociception and pain. Livrizzi, Chang-Weinberg *et al.* showed that vlPAG projections to RVM — and that continued opioid activity in this pathway — are critical for both placebo and morphine analgesia. Human studies have identified placebo-induced opioid release in PAG, aMCC, and vmPFC, and fMRI signal changes in all these elements of this circuit including spinal cord, with placebo analgesia.

that medial prefrontal input to PAG is crucial for pharmacologically conditioned placebo analgesia, but morphine can activate the vPAG descending modulatory system by other means.

As with any good study, new insights lead to new questions. The study does a remarkable job at identifying mPFC→PAG pathways for conditioned analgesia that appear to be conserved across species. But a number of mysteries remain. Some are focused on the brain: the wider network that creates and consolidates placebo-related memories (cf.¹⁴) is largely unexplored. Other recent placebo studies identify different pathways, such as one from ACC→pontine nuclei→cerebellum¹⁵. Whether these are part of an interconnected system with the ACC→PAG→RVM pathway or divergent mechanisms for different contextual manipulations remains unclear.

Other unanswered questions concern the process of creating placebo effects. We don't yet understand the boundary

conditions that lead to placebo effects or their absence. Curiously, in Livrizzi, Chang-Weinberg *et al.*'s study, the association between place context and morphine only produces analgesia if there is also exposure to noxious stimuli. Context-drug associations are not enough, and there is some as yet unknown interaction with painful stimulation that makes placebo effects work. This parallels human studies showing that pain may be a precondition for placebo analgesia^{16,17}. Livrizzi and Chang-Weinberg's study also points to a temporal sequence of processing that remains opaque. Decreasing the wait between saline/context exposure and hot plate from 30 to 10 minutes abolished the placebo effect. Current theories do not explain why.

A related, fundamental mystery is why exactly a morphine-associated context should produce *less* pain. Predictive processes exist to allow organisms to *compensate* for exogenous perturbations and maintain allostasis, not mimic them¹⁸.

Anticipation of a large meal elicits anticipatory insulin release, which reduces blood glucose¹⁹. Threat of pain elicits anticipatory opioid release, which reduces nociception and enables defensive behavior (also via PAG)²⁰. Why shouldn't context associated with exogenous opioids reduce the need for endogenous analgesia and therefore *increase* pain in the absence of drug? One possibility is that morphine doesn't simply produce antinociception. It may instead license an internal *response* or high-level action schema — something like “it's OK to ignore the pain now” — that conditioned cues re-activate.

In sum, Livrizzi, Chang-Weinberg *et al.*'s study is a powerful example of how the stunning methodological precision of modern neuroscience is being brought to bear on the age-old question of how thinking changes feeling. In a time in which traditional context-free pharmacology is yielding few wins for pain and mental health disorders, this approach is a welcome one.

DECLARATION OF INTERESTS

The author declares no competing interests.

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Turtle evolution: Missing a missing link

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Turtles are unique among living vertebrates in that their body is protected by a shell, but what the first turtle looked like has remained controversial. A new study reappraises the status of *Eunotosaurus* — the iconic ‘missing link’ in turtle evolution — with surprising results.

In the famous Aesop fable, the slow-paced tortoise wins a race against an over-confident hare by slowly, but steadily, going for the finish line. This may also be true for the study of turtle evolution. Much

like turtles, the evolutionary relationships of hares and rabbits have been debated for over a century due to a puzzling anatomy and a rather uninformative fossil record, and both were only settled in the

era of molecular analyses¹. The history of turtle evolution has been intertwined with an enigmatic South African fossil, *Eunotosaurus africanus* (Figure 1), which is revisited in a recent study by Xavier